



ISARIC Data Platform Statistical Analysis Plan

ALL APPLICATIONS MUST BE COMPLETED AND SUBMITTED VIA: data@isaric.org

All data contributors who generated the data you are requesting will be invited to participate in the analysis by reviewing and providing input on the statistical analysis plan and the resulting publication. The findings from this work will be widely disseminated to inform patient care and public health policy, including submission to an international, peer-reviewed journal. ISARIC aims to recognise all the names of data contributors as cited collaborators in the resulting publication, in accordance with ISARIC's publication policy.

Part 1: General Information and Research team

1. Study Title

The title of the study should be clear, concise, and should accurately convey the primary purpose of the study.

Global Differences in Systemic Corticosteroid Treatment Pathways and Clinical Outcomes Among Hospitalized COVID-19 Patients Across Income-Level Settings

2. Study Version and Date

List the study version and date of the SAP.

Version	
Date (09/03/2026)	(09/03/2026)

3. Analysis Lead

List the full name, institution and country, and e-mail address for correspondence of the Analysis Lead.

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4. Other investigators/Working Group Members

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5. Members requiring access to data

List the full name, institution and country, and e-mail address for correspondence of all the Members of the research team who will be analysing the data. Anyone who will have access to data must have signed an Oxford Researcher Agreement.

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Part 2: Research Plan

6. Background, Research Objectives, and Scientific Value

This section should explain the rationale for the research aim and objectives and support this with relevant information from scientific or other literature. All supporting statements should be referenced in this section with the full reference included in the "Reference" section of the form.

Since the emergence of COVID-19, systemic corticosteroids have become a cornerstone in the management of hospitalised patients with moderate to severe disease. Following the

results of large, randomised trials, particularly the RECOVERY trial, corticosteroids were widely adopted as standard therapy for patients requiring oxygen support. (1,2) Despite global guideline recommendations, the implementation of corticosteroid therapy varies substantially across healthcare systems, particularly between high-income countries (HICs) and low- and middle-income countries (LMICs). Differences may occur in:

- frequency of steroid use
- timing of initiation
- duration of therapy
- integration within broader treatment pathways

Healthcare resource availability, local treatment protocols, drug accessibility, and healthcare infrastructure may influence these practices. (3)

Understanding how corticosteroid treatment pathways differ across income-level settings may provide important insights into global health disparities and treatment implementation during pandemics.

Large international datasets, such as the ISARIC COVID-19 Clinical Characterisation dataset, provide an opportunity to examine these treatment patterns across diverse healthcare environments. (4)

This study aims to compare differences in corticosteroid treatment pathways between HIC and LMIC hospitals and assess their association with clinical outcomes.

7. Proposed Target Population

This section should describe the specific target population that the study will focus on.

Study Population

Inclusion Criteria

- Adult patients (≥ 18 years)
- Laboratory-confirmed SARS-CoV-2 infection
- Hospital admission for COVID-19
- Available data regarding corticosteroid administration and outcomes

Exclusion Criteria

- Missing key treatment variables (corticosteroid use, timing, or duration)
- Missing outcome data
- Incomplete hospitalization records

8. Clinical Questions

This section should outline the key clinical questions that the statistical analysis aims to address. These questions should be clearly defined and directly related to the objectives of the study.

Primary Study Question

Do hospitalized COVID-19 patients in low- and middle-income countries receive systemic corticosteroid therapy used for COVID-19 management (e.g., dexamethasone or equivalent corticosteroids such as methylprednisolone or prednisolone) with different frequency, timing, and duration compared with patients in high-income countries, and how are these differences associated with clinical outcomes?

Secondary Study Questions

1. Are corticosteroids initiated earlier or later in hospitalized COVID-19 patients in LMICs compared with HICs? when assessed both from time of hospital admission and from symptom onset?
2. Does the duration of corticosteroid therapy differ between LMIC and HIC hospitals?
3. Are other systemic treatments (e.g., antivirals, immunomodulators, anticoagulation) used differently between income-level settings?
4. Are differences in treatment pathways associated with:
 - ICU admission
 - hospital length of stay
 - in-hospital mortality
 - systemic complications, including secondary infections (bacterial or fungal), sepsis, thrombotic events, and other major in-hospital complications, where data is available?
 - Age & Sex?

9. Statistical Analysis and Methodology

This section should detail the planned data analysis. Specify the analytical methods to be used, such as calculating measures of central tendency (e.g., mean, median), and any statistical tests that will be conducted. Additionally, outline the strategies that will be implemented to handle potential issues related to missing data.

This study will be conducted as a retrospective multinational observational cohort study using anonymized patient-level data from the ISARIC COVID-19 Clinical Characterisation dataset.(7).

Patients hospitalized with laboratory-confirmed COVID-19 will be included and categorized according to World Bank country income classification, dividing hospitals into:

- High-income countries (HIC)
- Low- and middle-income countries (LMIC)

The study will compare systemic corticosteroid therapy used for COVID-19 management (e.g., dexamethasone or equivalent corticosteroids such as methylprednisolone or prednisolone) including treatment frequency, timing, and duration, across these income-level settings.(8)

Study Population

Eligible patients will include adults aged 18 years or older hospitalized with confirmed SARS-CoV-2 infection.

Patients with missing key variables related to corticosteroid administration or clinical outcomes will be excluded from the primary analysis.(5)

Variables

Exposure Variable: The primary exposure variable is country income level, categorized as: **(HIC) and (LMIC)**

Country classification will follow World Bank income group definitions.

Treatment Variables: The main treatment variable is systemic corticosteroids therapy.

The following steroid-related variables will be analyzed:

- steroid administration :
Dexamethasone (yes/no)
Methylprednisolone (yes/no)
Prednisolone (yes/no)

- timing of steroid initiation (days from hospital admission to first dose)
- duration of corticosteroid therapy (total days administered)

Additional systemic treatments will also be evaluated to characterize treatment pathways, including:

- antiviral therapy
- immunomodulatory therapy
- anticoagulation therapy

Severity Indicators

Baseline severity indicators will be assessed using available clinical indicators (e.g., oxygen saturation, respiratory rate, level of consciousness, ICU admission), and, where feasible, validated severity scores such as SOFA. These variables include:

- ICU admission
- oxygen saturation at admission
- respiratory rate
- level of consciousness (Glasgow Coma Scale GCS)

Outcome Measures

Do differences in corticosteroid treatment between HIC and LMIC associated with differences in patient outcomes such as mortality, ICU admission, and complications?

Secondary outcomes include:

- ICU admission
- hospital length of stay
- adverse events

Descriptive Analysis

Baseline patient characteristics will be summarized using:

means and standard deviations for normally distributed continuous variables medians and interquartile ranges for non-normally distributed variables frequencies and percentages for categorical variables

Comparisons between HIC and LMIC patient groups will be conducted.

Comparative Analysis

- Treatment Use
- Differences in corticosteroid use between income-level groups will be evaluated using:
- Chi-square tests for categorical variables.
- Treatment Timing and Duration
- Differences in steroid timing and duration will be assessed using:
- Independent t-tests for normally distributed data
- Mann–Whitney U tests for non-normally distributed data

Stratified Analysis

To account for differences in disease severity, analyses will be stratified using available clinical indicators reflecting physiological status, including respiratory function (e.g., oxygen saturation or need for oxygen support), level of consciousness, and other vital signs. Where sufficient data are available, validated severity scores such as SOFA will be explored.

Additional analyses will be performed stratified by sex, and age-specific analyses in females will be conducted using a cutoff of 55 years to account for potential hormonal differences and stratifying countries into high-, middle-, and low-income groups. Sensitivity analyses by geographic regions will also be considered where appropriate. ICU admission will not be used as a primary stratification variable due to potential variability in resource availability across settings.

This stratification will allow comparison of treatment patterns within similar severity groups.

Multivariable Regression Analysis

Multivariable regression models will be used to evaluate the association between corticosteroid treatment patterns and clinical outcomes, with time-to-event analyses using Cox proportional hazards models considered where appropriate to account for the timing of treatment and outcomes, while adjusting for potential confounding variables.⁽⁶⁾ A directed acyclic graph (DAG) will be used to identify relevant confounders and guide variable selection for the regression analysis.

- Logistic Regression
- Logistic regression models will be used for binary outcomes, including:
- ICU admission
- in-hospital mortality
- Predictor variables will include:
- country income level
- steroid use
- steroid timing
- age

- sex
- comorbidities
- severity indicators (SOFA)
- Linear Regression

Linear regression models will be used for continuous outcomes, including:

- hospital length of stay
- duration of corticosteroid therapy
- Missing Data

The proportion of missing data will be evaluated for all variables.

If appropriate, multiple imputation methods may be applied to handle missing data.

Sensitivity analyses will be conducted to assess the impact of missing values on study results.

Statistical Significance

All statistical tests will be two-sided, and a p-value <0.001 will be considered statistically significant. Adjustment for multiple comparisons will be performed using a false discovery rate (FDR) approach where appropriate.

Analyses will be conducted using appropriate statistical software such as R, Stata, or SPSS.

10. References

This section should list all sources cited throughout the statistical analysis plan.

1. RECOVERY Collaborative Group. Dexamethasone in hospitalized patients with Covid-19. N Engl J Med. 2021;384(8):693-704. doi:10.1056/NEJMoa2021436.

2. WHO Rapid Evidence Appraisal for COVID-19 Therapies (REACT) Working Group, Sterne JAC, Murthy S, Diaz JV, Slutsky AS, Villar J, et al. Association between administration of systemic corticosteroids and mortality among critically ill patients with COVID-19: a meta-analysis. JAMA. 2020;324(13):1330-1341. doi:10.1001/jama.2020.17023.
3. World Health Organization. Therapeutics and COVID-19: living guideline. Geneva: World Health Organization; 2023.
4. Docherty AB, Harrison EM, Green CA, Hardwick HE, Pius R, Norman L, et al. Features of 20 133 UK patients in hospital with COVID-19 using the ISARIC WHO Clinical Characterisation Protocol: prospective observational cohort study. BMJ. 2020;369:m1985. doi:10.1136/bmj.m1985.
5. ISARIC Clinical Characterisation Group. Global clinical platform for COVID-19 clinical characterization: development, standardization and data collection. BMJ Glob Health. 2020;5(12):e003350.
6. Harrell FE. Regression Modeling Strategies: With Applications to Linear Models, Logistic Regression, and Survival Analysis. 2nd ed. New York: Springer; 2015.
7. Vittinghoff E, Glidden DV, Shiboski SC, McCulloch CE. Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models. New York: Springer; 2012.
8. World Bank. World Bank country and lending groups. Washington (DC): World Bank; 2024. Available from: <https://datahelpdesk.worldbank.org>.

11. Appendices

Please provide all appendices related to this research.

Appendix A. Variable Definitions

Variable	Definition
Corticosteroid use	Binary variable indicating whether any systemic corticosteroid (e.g., dexamethasone, methylprednisolone, hydrocortisone) was administered during hospitalization, derived from individual drug-specific variables.
Steroid timing	Number of days from hospital admission to the first administration of systemic corticosteroids.
Steroid duration	Total number of days during which systemic corticosteroids were administered.

ICU admission	Admission to an intensive care unit at any point during hospitalization.
Hospital length of stay	Number of days from hospital admission to discharge or death.
In-hospital mortality	Death occurring during hospitalization for COVID-19.

Appendix B. Country Income Classification

Countries will be categorized according to the World Bank income classification (2024) into the following groups:

- High-income countries (HIC)
- Low- and middle-income countries (LMIC)

Country income classification will be determined using the World Bank Country and Lending Groups database.

Appendix C. Multivariable Logistic Regression Model

Multivariable logistic regression models will be used to evaluate the association between corticosteroid treatment patterns and in-hospital mortality.

The regression model will adjust for potential confounding variables including:

- age
- sex
- comorbidities
- ICU admission
- baseline disease severity indicators (e.g., oxygen saturation, respiratory rate, level of consciousness)

Adjusted odds ratios with 95% confidence intervals will be reported.